LC Optics and Waves Lecture Notes

Year 1 Semester 1

Ash Stewart

MSci Physics with Particle Physics and Cosmology

School of Physics and Astronomy
University of Birmingham

Table of Lectures

Lecture 1: Intro to Waves and SHM Recap **2**

Lecture 2: Wave Functions **4**

Lecture 3: Generalised Wavefunctions **6**

Lecture 4: Waves at Boundaries **10**

Lecture 5: Wave Applications and Introduction to Standing Waves **16**

Lecture 6: Standing Waves 2 Electric Boogaloo **20**

Lecture 7: Energy and Power of Waves **22**

Lecture 8: EM Standing Waves and Lasers **26**

Lecture 9: Doppler Shift, Shockwaves and Optics I **29**

Lecture 10: Optics II **34**

Lecture 11: Reflection and Refraction Examples I **38**

Lecture 12: Reflection and Refraction Examples II **43**

Lecture 13: Image Formation **44**

Wed 01 Oct 2025 11:00

Lecture 1 - Intro to Waves and SHM Recap

0.1 Course Objectives

- Have a sound understanding of basic wave properties
- Have a basic understanding of interference effects, inc diffraction
- Be able to use simple geometric optics and understand the fundamentals of optical instruments.

0.2 Recommended Textbooks

1. University Physics, Young and Freedman (Ch 15, 16 for Waves, Ch 33-36 for Optics)
2. Physics for Scientists and Engineers (Ch 20, 21 for Waves, Ch 22-24 for Optics)
3. 5e, Tipler and Mosca, (Ch 15, 16 for Waves, 31-33 for Optics)
4. Fundamentals of Optics, Jenkins and White
5. Optics, Hecht and Zajac

What is a wave? Waves occur when a system is disturbed from equilibrium and the disturbance can travel from one region to another region. Waves carry energy, but do not move mass. The course aim is to derive basic equations for describing waves, and learn their physical properties.

0.3 Periodic Motion

Waves are very linked to periodic motion. Therefore we recap periodic motion first.

It has these characteristics:

- A period, T (the time for one cycle)
- A frequency, f, the number of cycles per unit time ($f = \frac{1}{T}$)
- An amplitude, A, the maximum displacement from equilibrium.

Periodic motion continues due to the restoring force. When an object is displaced from equilibrium, the restoring force acts back towards the equi point. The object reaches equi with a non-zero speed, so the motion continues past the equi point and continues forever.

0.4 Energy

Periodic motion is an exchange between potential and kinetic energy, with no energy loss. Energy is conserved.

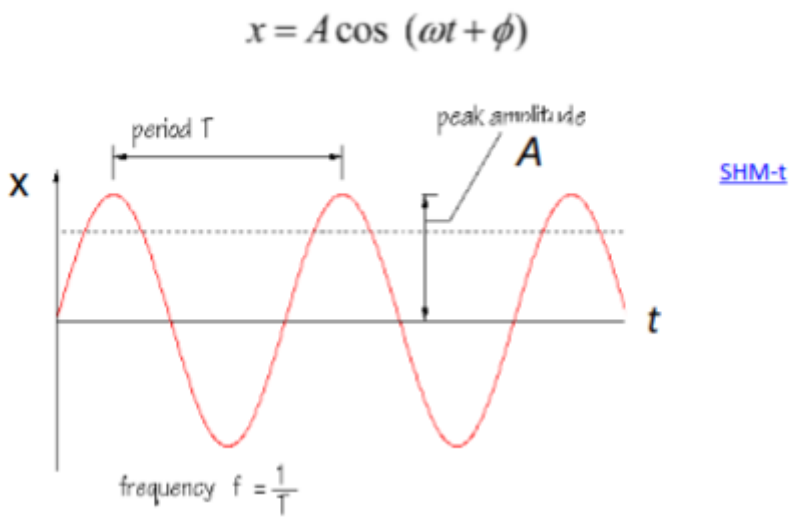
0.5 Simple Harmonic Motion

If the restoring force is directly proportional to the displacement $F = -kx$, then the periodic motion becomes Simple Harmonic Motion and the object is called a harmonic oscillator.

In a single dimension, displacement is given by:

$$x = A\cos(\omega t + \phi)$$

Where $\omega = 2\pi f$ is the angular velocity, and ϕ is the phase angle. In cases like this, where the phase angle is 90 deg we can simplify to $x = -A\sin(\omega t)$



SHM-t

Figure 1.1: A Phase Angle of 90

0.6 More SHM Equations

Velocity

$$v_x = \frac{dx}{dt} = -\omega A \sin(\omega t + \phi)$$

Acceleration

$$a_x = \frac{dv_x}{dt} = \frac{d^2x}{dt^2} = -\omega^2 A \cos(\omega t + \phi)$$

Both properties are signed to indicate direction, as they are both vectors.

Thu 02 Oct 2025 13:00

Lecture 2 - Wave Functions

0.1 Sine Waves

Mechanical Waves A mechanical wave is a disturbance through a medium. It's formed of a single wave pulse or a periodic wave.

Mechanical Waves have the following properties:

1. Transverse: Where displacement of the medium is perpendicular to the direction of propogation.
2. Longitudinal, displaccement of hte medium is in the same direction as propogation.
3. Propogation depends on the medium the wave moves through (i.e. density, rigidity)
4. The medium does not travel with the wave.
5. Waves have a magnitude and a direction.
6. The disturbance travels with a known exact speed.
7. Waves transport energy but not matter throughout the medium.

0.2 Wave Functions

We want to define a wave function in terms of two variables, x and t. In any given moment, if we consider a single point on the wave (i.e. $t = 0$), and wait a short while, the wave will have travelled to some $t = t_1 > 0$.

In order to quantify displacement, we therefore want to specify both the time, and the displacement. This will let us find the wave speed, acceleration and the (new) wave number.

We are also able to talk about the velocity and acceleration of individual particles on the wave.

0.3 Wave Function for a Sine Wave

Consider a sine wave. We want to find a wave funtion in the form $y(x, t)$. Consider the particle at $x = 0$.

We can express the wave function at this point as $y(x = 0, t) = A \cos \omega t$. However we want to expand this to any general point. Now consider a point (2) which is one wavelength away. We know the behaviour of particle 1 is mirrored by particle 2 (with a time lag).

Since the string is initially at rest, it takes on period (T) for the propogation of the wave to reach point 2, therefore point 2 is lagging behind the motion of point 1. The wave equation is therefore (if particle two has $x = \lambda$) $y(x = \lambda, t) = A \cos(\omega t - 2\pi)$.

For arbitrary x, $y(x, t) = A \cos(\omega t - \frac{x}{\lambda} \cdot 2\pi)$ to account for this delay. This quantity is called the wave number:

$$\text{Wave Number: } k = \frac{2\pi}{\lambda}$$

So:

$$\begin{aligned} y(x, t) &= A \cos(\omega t - kx) \\ &= A \cos(kx - \omega t) \end{aligned}$$

Note the second step is possible as cos is an even function. k can also be signed to indicate direction: if $k > 0$, the wave travels in the positive x. If $k < 0$, the wave travels in the negative x direction. Again, $\omega = 2\pi f$

0.4 Displacement Stuff

Considering a point (starting at equi), the time taken for the particle on the sin wave to reach maximum displacement, minimum displacement and back takes the time period T. The speed of the wave is distance travelled over the time taken. We take the distance to be the wavelength λ, as we know the time by definition this takes is one time period T. Therefore wave speed v is:

$$v = \frac{\lambda}{T} = \lambda f$$

Since $\lambda = \frac{2\pi}{k}$ and $f = \frac{\omega}{2\pi}$ (as ω is defined as $\frac{2\pi}{T}$), we can also write:

$$v = \frac{2\pi}{k} \cdot \frac{\omega}{2\pi} = \frac{\omega}{k}$$

0.5 Particle Velocity

We can also determine the velocity of individual particles in the medium. We can use this to determine the acceleration.

We know that

$$y(x,t) = A\cos(kx - \omega t)$$

The vertical velocity v_y is therefore given by:

$$v_y = \frac{dy(x,t)}{dt}$$

Which is unhelpful (as we can't differentiate two variables at once), we can slightly cheat this by looking at purely a certain value of x, and therefore treating x as constant (to get a single variable derivative).

$$v_y = \frac{dy(x,t)}{dt}\Big|_{x=\text{const.}}$$

However this is notationally yucky, so we therefore use the notation:

$$\frac{\partial y(x,t)}{\partial t}$$

To represent the same thing. Finally (carrying out the partial derivative):

$$v_y(x,t) = \frac{\partial y(x,t)}{\partial t} = \omega A\sin(kx - \omega t)$$

0.6 Particle Acceleration

We can work out particle acceleration (transverse acceleration) by differentiating in the same manner again:

$$a_y(x,t) = \frac{\partial^2 y(x,t)}{\partial t^2} = \frac{\partial}{\partial t}\left(\frac{\partial y(x,t)}{\partial t}\right) = -\omega^2 A\cos(kx - \omega t) = -\omega^2 y(x,t).$$

Wed 08 Oct 2025 11:00

Lecture 3 - Generalised Wavefunctions

Recap: For a sine wave, the wavefunction is:

$$y(x,t) = A\cos(kx - \omega t)$$

Where k is the wave number, $k = \frac{2\pi}{\lambda}$ and omega is $\frac{2\pi}{T} = 2\pi f$

1 More Wavefunctions

What is the general form of a triangular shaped wave? What about a square stepped wave? TODO: Diagram

We know that each particle (i.e. along a string) copies the motion of its immediate left-hand neighbour (for a particle moving in the positive x), with a time delay proportional to their distance. Every wave is describable $y(x,t)$ wave function, and every mechanical wave relies on a medium (i.e. a piece of string or water, concrete etc) to travel.

Ideally, we want to be able to find a general form of a wave function.

In a single fixed instant of time, the moving wave pulse is stationary, so purely a function of $y = f(x)$. We want a wave function where we can input any value of t, so we need a moving frame of reference. We define this frame of reference as O' (for the origin) and x', y' axes. This frame of reference moves with the wave pulse and at the same speed, therefore y' is a function of x' only, independant of speed.

1.1 New System

x is the distance from the origin O to the relevant point, while x' is the distance from O'.

$$x = x' + vt$$

$$x' = x - vt$$

$$y' = f(x') = f(x - vt)$$

However, as the wave is moving purely in one direction (along x), $y = y'$, so:

$$y = f(x - vt)$$

1.2 Back to Basics

Going back to:

$$\begin{aligned} y(x,t) &= A\cos(kx - \omega t) \\ &= A\cos\left[k\left(x - \frac{\omega}{k}t\right)\right] \\ &= A\cos\left[k\left(x - vt\right)\right] \end{aligned}$$

(Note, this is true for a wave in the positive x, for a wave moving in the negative x this would be x + vt)

1.3 Equivalent Rrepresentations

There are some equivalent representations for a sine wave:

$$y(x,t) = A\cos\left(2\pi f\frac{x}{v} - \omega t\right) = A\cos\left(2\pi\left[\frac{x}{\lambda} - \frac{t}{T}\right]\right) = TODO$$

2 More Differentiation

We've already looked at differentiating with respect to t, but what about x? This would give us the slope of the string at that point:

$$\frac{\partial y(x,t)}{\partial x}$$

And the curvature of the string:

$$\frac{\partial^2 y(x,t)}{\partial x^2} = -k^2 A\cos(kx - \omega t) = -k^2 y(x,t)$$

Note the similarities here with the equation for transverse acceleration:

$$(1)\frac{\partial^2 y(x,t)}{\partial t^2} = -\omega^2 y(x,t)$$

$$(2)\frac{\partial^2 y(x,t)}{\partial x^2} = -k^2 y(x,t)$$

Dividing (1) by (2):

$$\frac{\partial^2 y(x,t)/\partial t^2}{\partial^2 y(x,t)/\partial x^2} = \frac{\omega^2}{k^2} = v^2$$

Therefore:

$$\frac{\partial^2 y(x,t)}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y(x,t)}{\partial t^2}$$

We call this the 'wave equation' as every wave function y(x, t) must satisfy it, regardless of whether or not it is periodic or its direction of travel. If y(x, t) does not satisfy this, it is not a wave function.

2.1 An Example

$$y(x,t) = \frac{x^3 - vt^2}{e^t}$$

TODO, the example in our own time

3 Wave Equation for a String

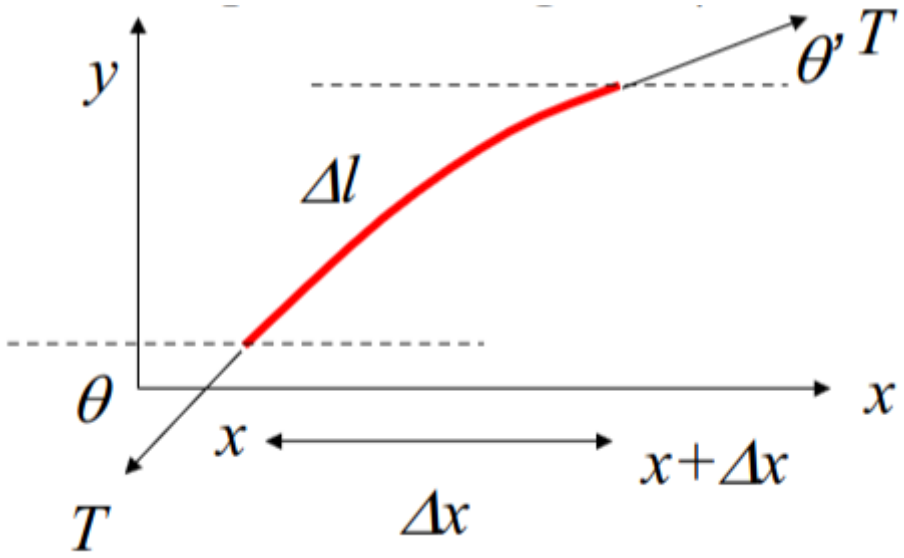


Figure 3.1: A snapshot of the wave.

Lets say we have some string, suspended horizontally under tension. We generate a single wave pulse and allow it to propogate down the string.

We assume the string is 1D, under tension T (constant throughout) and has mass per unit length μ.

Consider a small segment of string of length ΔL from x to $x+\Delta x$. This string makes angle θ with the horizontal at the bottom of the string, and angle θ' with the horizontal at the top of the string.

Net force (transverse in y) is:

$$F_y = T\sin\theta' - T\sin\theta$$

And using the small angle approx $\sin\theta \approx \tan\theta = \frac{dy}{dx}$:

$$F_y = T\left(\frac{dy}{dx}\bigg|_{x+\Delta x} - \frac{dy}{dx}\bigg|_x\right)$$

Using differentiation by first principles:

$$\frac{dy}{dx} = \lim_{\Delta x\to 0}\frac{f(x+\Delta x)-f(x)}{\Delta x}$$

We can say:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \lim_{\Delta x\to 0}\frac{\frac{dy}{dx}\Big|_{x+\Delta x} - \frac{dy}{dx}\Big|_x}{\Delta x}$$

Therefore:

$$F_y = T\frac{d^2y}{dx}\Delta x$$

The mass of this section of string is $\mu\Delta x$, and considering the acceleration in the y direction we can plug into $F = ma$ to get:

$$F = ma$$

$$T\frac{d^2y}{dx^2}\Delta x = \mu\Delta x\frac{d^2y}{dt^2}$$

$$\frac{d^2y}{dx^2} = \frac{\mu}{T}\frac{d^2y}{dt^2}$$

The LHS is the rate of change of the string's gradient, which as mentioned is the curvature of the string. The RHS includes the transverse acceleration of the string, therefore acceleration is proportional to curvature. To evaluate this at a fixed time/position we should write:

$$\frac{\partial^2y}{\partial x^2} = \frac{\mu}{T}\frac{\partial^2y}{\partial t^2}$$

And comparing to the wave equation we get:

$$\frac{1}{v^2}\frac{\partial^2y}{\partial t^2} = \frac{\mu}{T}\frac{\partial^2y}{\partial t^2}$$

$$\frac{1}{v^2} = \frac{\mu}{T}$$

$$v = \sqrt{\frac{T}{\mu}}$$

So a wave travels faster under a higher tension with a lower mass per unit length.

Thu 09 Oct 2025 13:00

Lecture 4 - Waves at Boundaries

1 Recap

We previously derived:

$$v = \sqrt{\frac{T}{\mu}}$$

And we also have:

$$v = \frac{\omega}{k}$$

The former is useful explictly for a wave travelling over a string, while the latter is applicable to the movement of any wave. On a string, a higher tension yields a higher restoring force and therefore a higher speed, while a higher mass per unit length gives a higher mass for some arbitrary length of string, therefore a lower acceleration and lower speed.

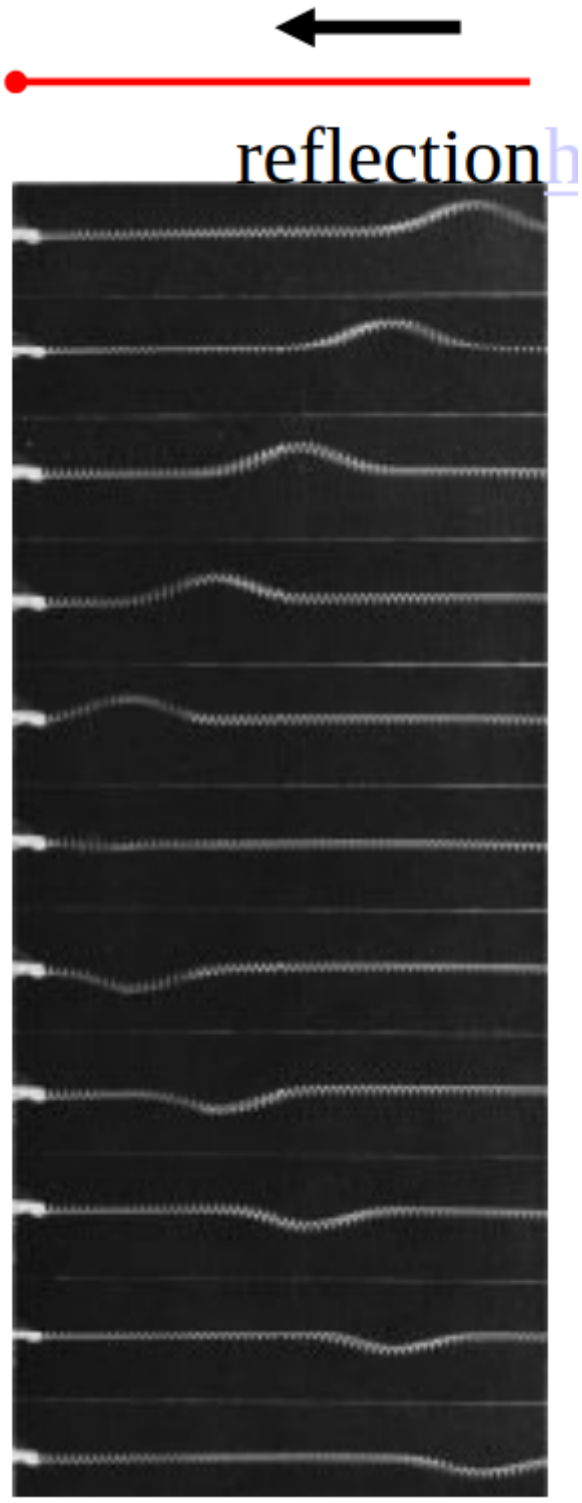
2 A Quick Interlude

The speed of a mechanical wave has the general form:

$$v = \sqrt{\frac{\text{Restoring force returning to equilibrium}}{\text{Inertia resisting return to equilibrium}}}$$

3 Reflection

When a wave hits a fixed boundary, it is reflected and inverted. Lets consider a case where a string is fixed on the LHS and is reflected back:


reflection

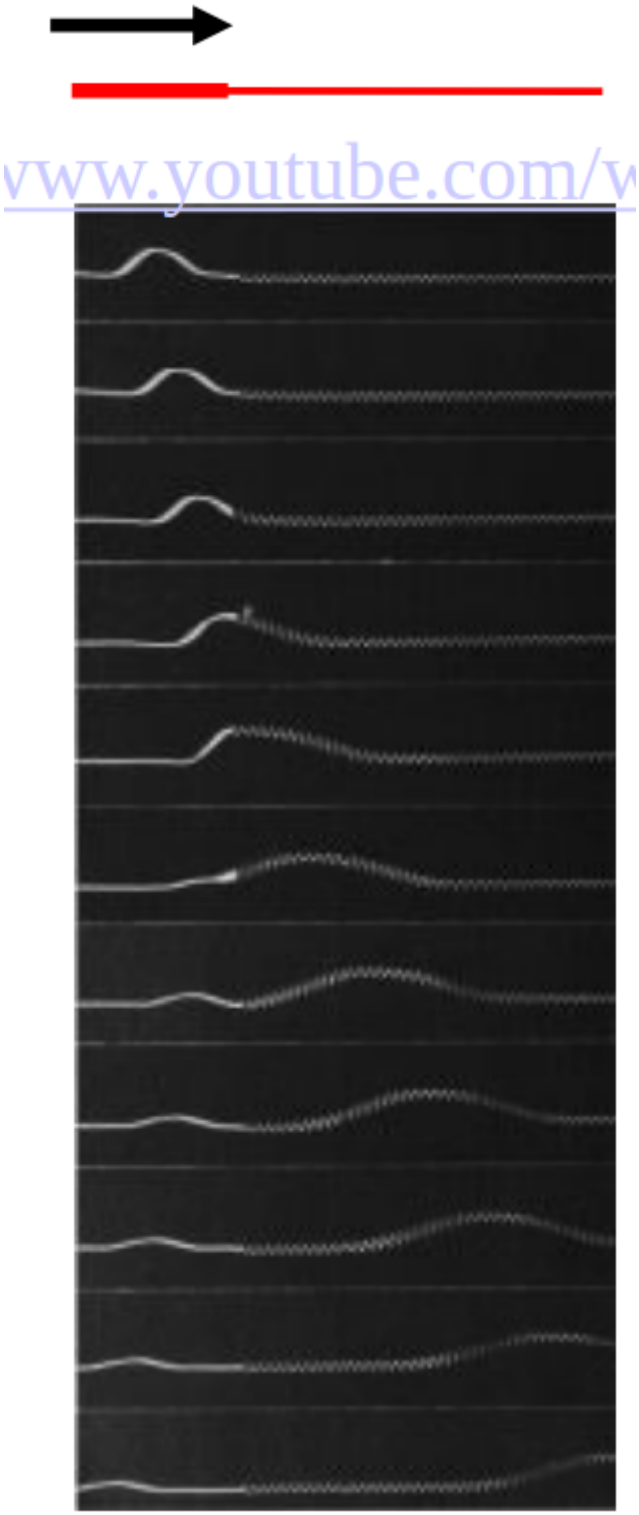


Figure 4.1: Or with two strings, going from thick to thin:

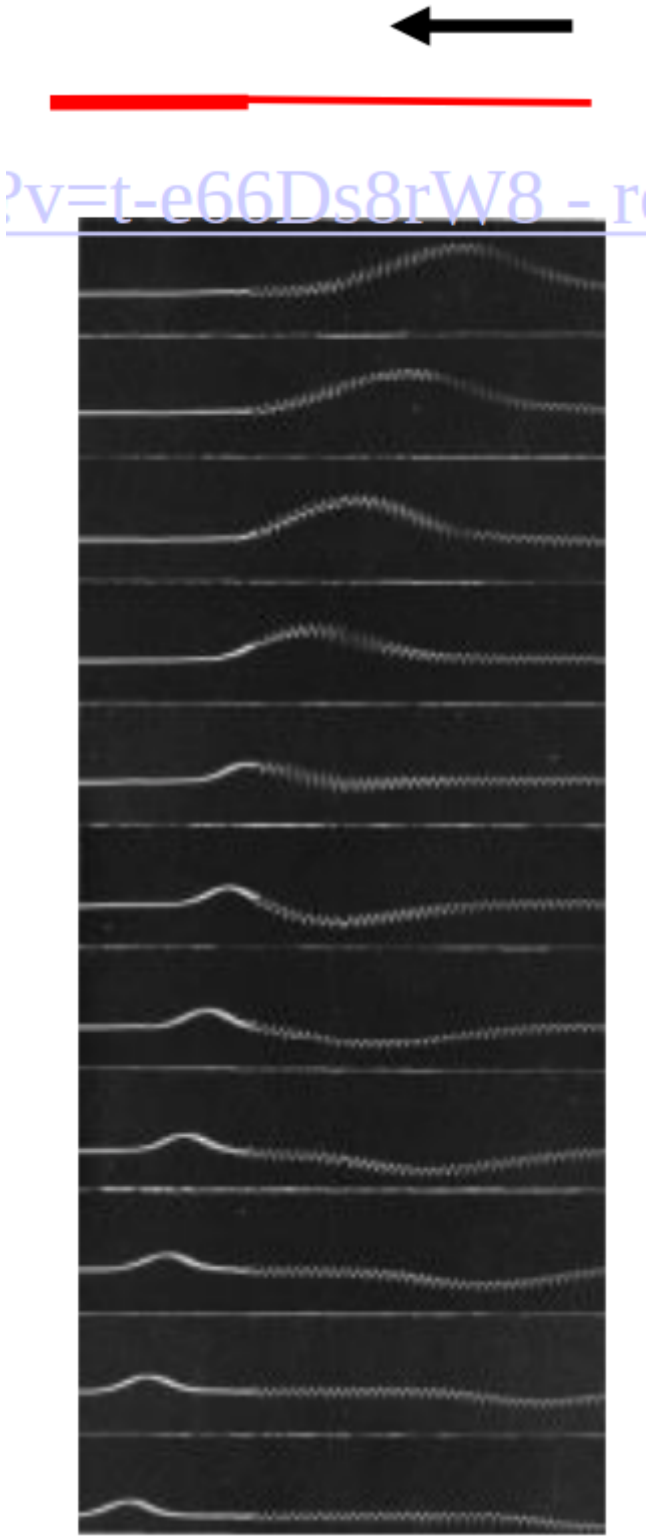


Figure 4.2: Or from thin to thick.

4 Waves Interacting at Boundaries

Say we have two pieces of string connected to each other, one thin string with mass per unit length μ_1 and a thicker string with μ_2 (both under the same tension, T). If a wave pulse is passed along from the thin string to the thicker string, whath happens at the point of connection, P?

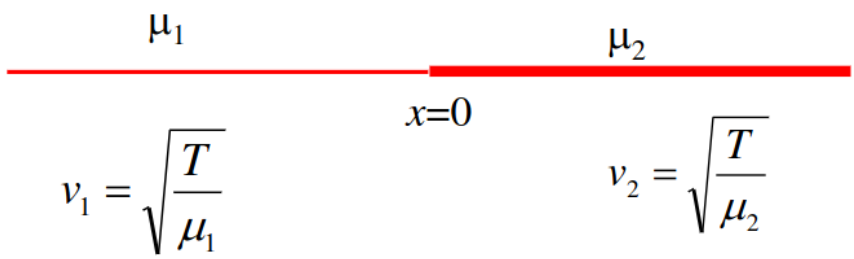


Figure 4.3: The two connected strings.

Consider a travelling wave coming from the left:

$$y_1 = A\cos(k_1 x - \omega_1 t)$$

$$y_2 = B\cos(k_2 x - \omega_2 t)$$

At the point of connection, we'll define this as $x = 0$. Here (as the string is not broken, so it must be connected):

$$y_1 = y_2 \tag{4.1}$$

Also, force is finite, therefore curvature must be finite (as force is proportional to curvature). Therefore we cannot have any discontinuities in curvature.

$$\frac{\partial y_1}{\partial x_1} = \frac{\partial y_2}{\partial x_2} \tag{4.2}$$

Disregarding non-linear effects, so assuming that the frequency with which the waves travel down the string is the same for both parts of the string, $\omega_1 = \omega_2 = \omega$.

From (1) and noting $x = 0$:

$$A\cos(-\omega t) = B\cos(\omega t) \tag{4.3}$$

And from (2) with the same note:

$$-k_1 A\sin -\omega t = -k_2 B\sin -\omega t \tag{4.4}$$

However this suggests that $A = B = 0$. This is technically a solution, but not really - it doesn't represent an actual wave (just two flat lines with no amplitude ever)

4.1 Including Reflection

The previous case did not work as we disregarded reflection at the wave boundary, P. Lets add an extra wave (the C term) to represent the reflection back into the light string:

$$y_1 = A\cos(k_1 x - \omega t) + C\cos(k_1 x + \omega t) = B\cos(k_2 x - \omega t)$$

From y1 = y2 at x = 0 we get:

$$y_1 = A\cos(-\omega t) + C\cos(-\omega t) = B\cos(-\omega t)$$

So: $A + C = B$

From (2) we get:

$$-k_1 A\sin(-\omega t) - k_1 C\sin(\omega t) = -k_2 B\sin(-\omega t)$$

Which has solutions:

$$B = \frac{2k_1}{k_1 + k_2}A$$

$$C = \frac{k_1 - k_2}{k_1 + k_2}A$$

We have the incident wave:

$$y_1 = Acos(k_1 x - \omega t)$$

The transmitted wave:

$$y_2 = B\cos(k_2 x - \omega t) = \frac{2k_1}{k_1 + k_2} A\cos(k_2 x - \omega t)$$

And lastly the reflected wave:

$$y_3 = C\cos(k_1 x + \omega t) = \frac{k_1 - k_2}{k_1 + k_2} A\cos(k_1 x - \omega t)$$

We note:

k_1 is the wave number in the medium where the incident wave comes from.

k_2 is the wave number in the medium where the transmitted wave goes into.

4.2 Example

If the wave comes in from the left, then $\mu_2 > \mu_1$ per example diagram, then $k_2 > k_1$, and C is negative.

Note that if the single pulse as a postive y amplitude, then the transmitted pulse in the heavier string will also have a positive y amplitude, but will have a smaller magnitude. The reflected wave will have a negative y-amplitude as the reflection inverts it.

If the wave comes from the right (from thick to thin), then $\mu_1 > \mu_2$, $k_1 > k_2$ and C is positive.

Wed 15 Oct 2025 11:00

Lecture 5 - Wave Applications and Introduction to Standing Waves

1 Wave Applications

Reflection and transmission have some very important real word applications:

- Fibre Optics.
- Vision and Photography.
- X-Ray Imaging.
- Sonar Imaging.
- Ultrasound Imaging
- Police Speed Checks.

We will look at a few in a bit more detail.

1.1 Electron Microscopes

We have two types:

- Scanning Electron Microscope (SEM)
  - We take some sample, and generate a very narrow beam of electrons. Some of these electrons will scatter off the sample back up. We take a detector, "scan" it around many angles and detect angles of scattering, to build a picture of the sample's surface.
- Transmission Electron Microscope (TEM)
  - We take a very thin sample, fire the electron beam at it. Some of the electron beam will pass through the sample, where we can detect it on the other side. Areas of the sample which are thinner act as being more transparent, with a greater rate of transmission. We can measure this difference.

1.2 Solar Cells

Consider a solar cell with light incident on it at some angle. We want to maximise absorption of light and minimise reflection, to maximise the energy efficiency of the cell.

1.3 Ultrasound Imaging

So far, we have considered a wave to only be able to transmit or reflect at a boundary. What we have not considered yet is attenuation. This is where a wave passing through a medium loses some energy to the medium. As the wave propagates, it loses energy and hence decreases in amplitude.

For example, a sound wave propagating though a medium causes particles in the medium to oscillate. This causes a heating effect on the medium, where some of the wave energy is lost thermally resulting in a decrease in amplitude (volume).

Ultrasound is sound with very high frequencies, compared to audible sound:

$$f_{\text{ultrasound}} : 2 - 20\text{MHz}$$

$$f_{\text{sound}} : 20\text{Hz} - 20\text{kHz}$$

Wave speed is given by the below, where ρ is density and B is bulk modulus (resistance to compression):

$$v = \sqrt{\frac{B}{\rho}}$$

This means:

- Ultrasound's wavelength in air (10MHz) is $33\mu m$.
- Sound's wavelength in air (at 300Hz) is $\approx 1m$

Ultrasound has a longer wavelength in fat (human tissue) of about $150\mu m$ as while fat does have a higher density, it has a much greater bulk modulus, and a greater B/ρ ratio.

Lets consider transmission of ultrasound at an air-fat boundary. Since the wavelength in air is smaller, and the density of air is much smaller, $C \approx A$, so $B \approx 0$ and there is (almost) zero transmission from air to fat.

This is why ultrasound gel is used, to ensure that there is transmission from air to gel and gel to fat, effectively bridging the gap.

2 Introduction to Standing Waves

Consider a string (of indefinite length) attached to the wall on the left. We send series of wave pulses (y_1) into the string from the right, which travel left and reflect. The reflected wave (y_2) travels back to the right.

If we continue sending a wave signal from $y_1(x,t)$ we would have a continuous wave signal back in the form of $y_2(x,t)$.

This causes the two waves to overlap and interfere with each other. We can say that the resultant amplitude for any point on the string is given by:

$$y = y_1(x,t) + y_2(x,t)$$

This works because the wave equation is a linear partial differential equation. If two functions y_1, y_2 are solutions to the wave equation, then this means their sum will be two (hence y is a valid final wave).

The incident wave is:

$$y_1 = -A\cos(kx + \omega t)$$

And the reflected wave is (assuming the two waves have equal amplitude, period and wavelength, with the second travelling in the opposite direction and being inverted due to the reflection):

$$y_2 = A\cos(kx - \omega t)$$

We add these together:

$$y_{\text{total}} = y_1 + y_2 = A\left[\cos(\underbrace{kx}_{A} - \underbrace{\omega t}_{B}) - \cos(kx + \omega t)\right]$$

$$\text{using: } \cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$y_{\text{total}} = A\left[-\cos(kx)\cos(\omega t) + \sin(kx)\sin(\omega t)\right] + A\left[\cos(kx)\cos(\omega t) + \sin(kx)\sin(\omega t)\right]$$

$$y_{\text{total}} = 2A\sin(kx)\sin(\omega t)$$

This gives us the general form of a sinusoidal standing wave function.

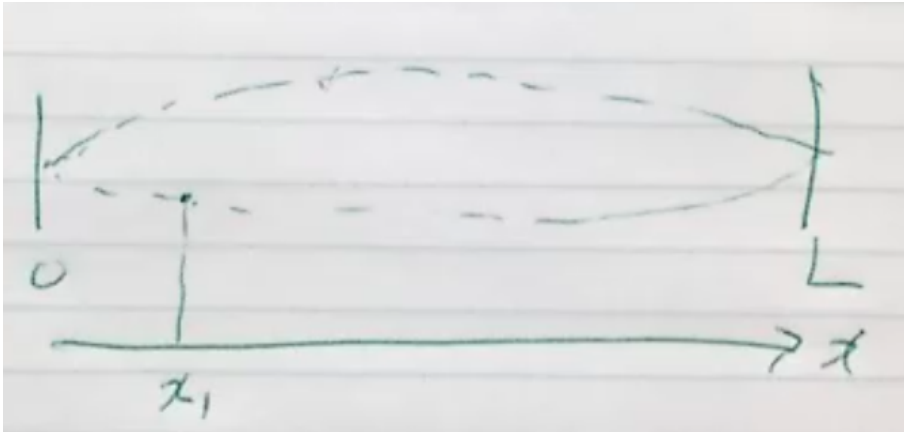


Figure 5.1

At some point $x = x_1$:

$$y_t(x = x_1, t) = \boxed{2A\sin(kx_1)}\sin(\omega t)$$

As x_1 is a constant, the boxed term is now another constant too. Let this constant be E.

$$y_t(x = x_1, t) = E\sin\omega t$$

This is the equation for a harmonic oscillator. This point of the string oscillates in simple harmonic motion with a maximum amplitude dependant on position.

If we consider a higher frequency wave, we may encounter points where $\sin kx = 0$. These are called nodes.

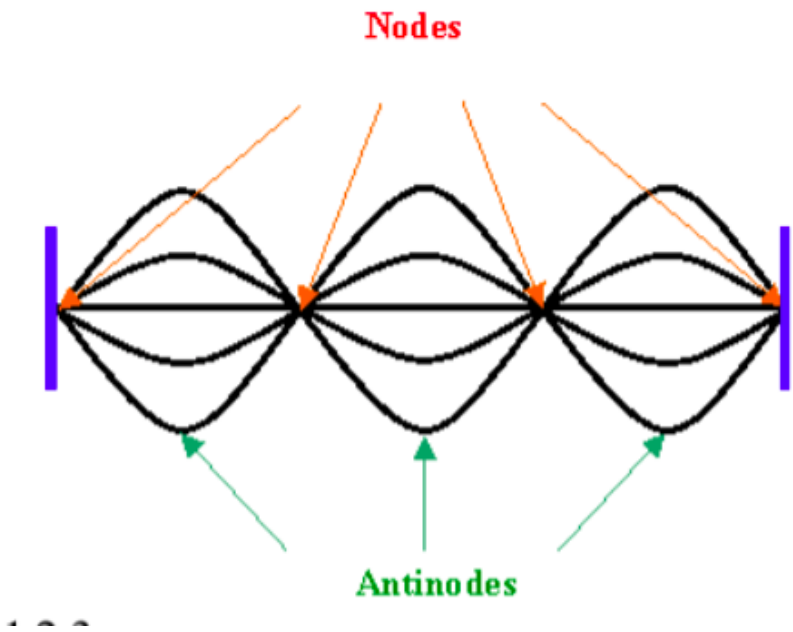


Figure 5.2

The amplitude at a node is always $y = 0$, regardless of time, as the sine term is zero regardless of time.

This wave does not propagate, it is "stationary".

3 Standing Waves and the Wave Equation

It's important to note that standing waves also satisfy the wave equation. Carrying out derivatives, we can get:

$$\frac{\partial y}{\partial x} = 2Ak\cos kx\sin\omega t$$

$$\frac{\partial^2 y}{\partial x^2} = 2A(-k^2)\sin kx \sin \omega t$$
$$\frac{\partial y}{\partial t} = 2A\omega \sin kx \cos \omega t$$
$$\frac{\partial^2 y}{\partial t^2} = 2A(-\omega^2)\sin kx \sin \omega t$$

And substituting into the wave equation:

$$\frac{\left(\frac{\partial^2 y}{\partial t^2}\right)}{\left(\frac{\partial^2 y}{\partial x^2}\right)} = \frac{2A(-\omega^2)\sin kx \sin \omega t}{2A(-k^2)\sin kx \sin \omega t} = \frac{\omega^2}{k^2} = v^2$$

$$\therefore \frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}$$

So the wave equation is satisfied.

Thu 16 Oct 2025 13:00

Lecture 6 - Standing Waves 2 Electric Boogaloo

1 Recap

$$\frac{\partial y}{\partial x} = 2Ak\cos(kx)\sin(\omega t)$$

$$\frac{\partial^2 y}{\partial x^2} = 2A(-k^2)\sin(kx)\sin(\omega t)$$

$$\frac{\partial y}{\partial t} = 2A\omega\sin(kx)\cos(\omega t)$$

$$\frac{\partial^2 y}{\partial t^2} = 2A(-\omega^2)\sin(kx)\sin(\omega t)$$

Therefore:

$$\frac{\frac{\partial^2 y}{\partial t^2}}{\frac{\partial^2 y}{\partial x^2}} = \frac{2A(-\omega^2)\sin(kx)\sin(\omega t)}{2A(-k^2)\sin(kx)\sin(\omega t)} = \frac{\omega^2}{k^2} = v^2$$

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}$$

Therefore a standing wave still obeys the wave equation, as it must.

2 Standing Wave Properties

2.1 Wavelength

Consider a horizontal string from $x = 0$ to $x = L$, with both ends fixed. We generate a sinusoidal wave pulse, which must satisfy:

$$y(x,t) = 2A\sin(kx)\sin(\omega t)$$

We know at $x = L$ and $x = 0$, $y = 0$ at all times as fixed at this point. Therefore:

$$kL = n\pi, (n \in \mathbb{N})$$

For $n = 1$, we have half a wavelength on the string:

$$\lambda = \frac{2L}{1} = 2L$$

And this has a general form: $\lambda = \frac{2L}{n}$.

2.2 Frequency

$$f_n = \frac{v}{\lambda_n} = \frac{v}{\left(\frac{2L}{n}\right)} = \frac{nv}{2L}$$

Crucially:

$$f_1 = \frac{v}{2L}$$

Is the first harmonic (or fundamental). $f_2 = 2f_1$ is the second harmonic, or first overtone, etc. All of these, f_n where $n \in \mathbb{N}$ are called "normal modes". For each normal mode, the corresponding frequency is called the resonant frequency (natural frequency of the system).

What happens if we try to create a standing wave where $\lambda \neq \frac{2L}{n}$? In short, we cannot. The system will reject any attempts to do so.

2.3 Energy

Energy is proportional to ω^2. Energy can only take certain discrete values (corresponding to $f_1, f_2, \ldots, f_n$), we find that the system has quantised possible values for energy.

To generate a wave with a higher frequency we either have to use a shorter L, or a higher v. A higher v is achieved by using a lighter string or placing the system under higher tension.

3 Sound Waves

3.1 Notation

Displacement of a sound wave is denoted:

$$s(x,t) = S_m \cos(kx - \omega t)$$

And pressure is given by:

$$\Delta P(x,t) = \Delta P_m \sin(kx - \omega t)$$

4 Different Boundary Conditions

The equations for standing waves given is only true for the boundary conditions of both ends fixed. If we vary these (for example left end fixed, right end not, wave initially travelling left) we get a different solution. For example, the first harmonic:

$$L = \frac{1}{4\lambda_1}$$

$$f_1 = \frac{v}{4L}$$

Where the left end forms a node (as required by boundary conditions) and the right end forms an antinode, as it is free to move. For the third harmonic:

$$L = \frac{3}{4\lambda_3}$$

$$f_3 = \frac{3v}{4L} = 3f_1$$

And fifth:

$$L = \frac{5}{4\lambda_5}$$

$$f_5 = \frac{5v}{4L} = 5f_1$$

Notably, this system cannot support even harmonics.

Thu 22 Oct 2025 11:00

Lecture 7 - Energy and Power of Waves

1 Kinetic Energy for a Sine Wave

Assuming a sine wave created by a harmonic oscillator. We can consider every particle on this string as acting as its own harmonic oscillator. We can determine the kinetic energy for one single harmonic oscillator (of an infinitesimally small length of string)

$$KE = \frac{1}{2}mv^2$$
$$(KE)_{\text{max}} = \frac{1}{2}mv_{\text{max}}^2$$

Mass is given by the density of the string per unit length multiplied by the length of the section of string: μdx, and the maximum velocity by ωA:

$$(KE)_{\text{max}} = \frac{1}{2}(\mu dx)(\omega A)^2$$

This gives us the kinetic energy of a single oscillator. We now want to get the total energy across a single wavelength, by integrating:

$$E = \int_0^\lambda \frac{1}{2}\omega^2 A^2 \mu dx$$
$$E = \frac{1}{2}\omega^2 A^2 \mu \int_0^\lambda dx$$
$$E = \frac{1}{2}\omega^2 A^2 \mu \lambda$$

Note that strictly speaking, at any point half the wave's energy is kinetic and half is potential. Instead of calculating potential energy, we consider a singular oscillator. In this single oscillator, the two quantities oscillate in opposition to each other, so max KE means zero PE. We can say $E_{\text{total}} = KE_{\text{max}}$ therefore, for a singular oscillator. This works out when we integrate across all oscillators.

And for power:

$$\text{Power} = \frac{\text{Energy}}{\text{Time}}$$
$$P = \frac{\frac{1}{2}(\omega A)^2 \mu \lambda}{T}$$
$$= \frac{1}{2}(\omega A)^2 \mu v$$

2 Standing Waves

Considering a standing wave with $\lambda = 2L$ (hence a single loop with two nodes at each boundary and one antinode in the middle):

$$y_t = y_1 + y_2$$
$$\implies (KE)_t = (KE)_1 + (KE)_2$$

Where y_1 and y_2 are identical except for their direction, but they carry the same kinetic energy. The KE of a standing wave is the sum of the KE of two waves that make it up.

Note: Normally, A is the amplitude of the travelling wave (hence 2A is the amplitude of the standing wave created by them), however it is sometimes ambiguous what is being referred to by A - some questions may give the standing wave amplitude and also denote this as A.

3 Interference

Superposition: When waves overlap in the same region, the resulting wave is the algebraic sum of waves (they interfere)

Consider two waves:

$$y_1 = A\cos(kx - \omega t) \qquad y_2 = A\cos(kx - \omega t + \delta)$$

Where both waves are travelling in the same direction with the same amplitude, but the second wave has some phase shift δ.

We can sum the waves as with standing waves:

$$y = A\cos(\underbrace{kx - \omega t}_{\alpha}) + A\cos(\underbrace{kx - \omega t + \delta}_{\beta})$$

$$\text{Using:} \quad \cos\alpha + \cos\beta = 2\cos\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}$$

$$y = 2A\cos\left(\frac{\delta}{2}\right)\cos\left(kx - \omega t + \frac{\delta}{2}\right)$$

So the amplitude of the new travelling (not standing) wave is dependant on delta, where amplitude is given by $2A\cos(\delta/2)$.

Here are three examples of phase difference where y_1 is the blue line, y_2 is green and y_{total} is red:

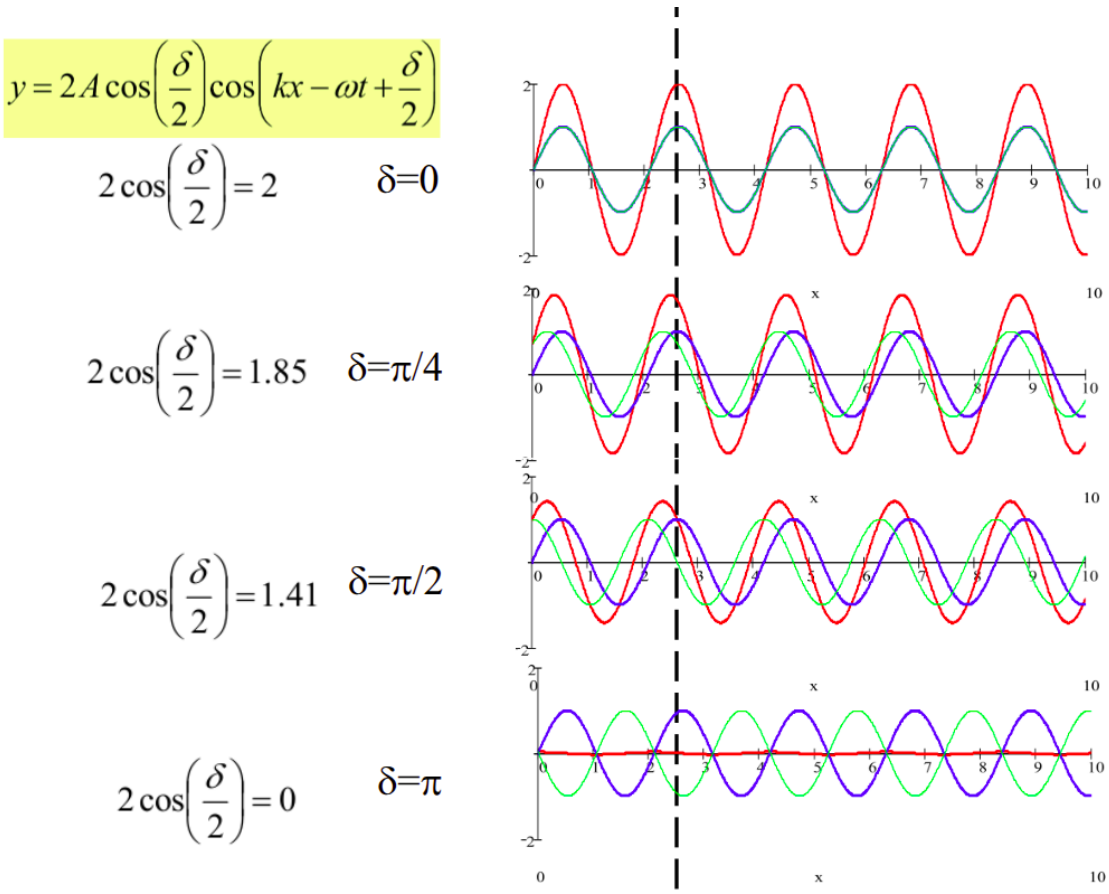


Figure 7.1

Given two arbitrary waves, differences in phase can arise from either a difference in angular frequency, and/or in x.

3.1 Example

Suppose we have two sound sources, S_1, S_2. They oscillate in phase and emit harmonic waves of equal frequency and wavelength. Lets consider some point P where the waves interact where the path differences are different (i.e. length from S1 to P $\neq$ length from S2 to P).

If there is an integer number of wavelengths in path difference, the interference is totally constructive, if path difference is equal to a half number of wavelengths (i.e. 1.5, 2.5, 3.5) then the interference is totally destructive.

From S_1:

$$y_1(l_1, t) = A\cos(kl_1 - \omega t)$$

From S_2:

$$y_2(l_2, t) = A\cos(kl_2 - \omega t)$$

Giving us a phase difference of:

$$\delta = (kl_1 - \omega t) - (kl_2 - \omega t) = k(l_1 - l_2)$$

$$\delta = \frac{2\pi}{\lambda}\Delta l$$

Hence the signal strength at point P is dependant on the path difference.

4 Beats

Lets consider interference from waves with slightly different frequencies (but the same amplitude). We again have two sources S_1 and S_2. We observe the resultant amplitude after these have interfered at some value of x (say $x = 0$ so the kx term is 0), where both sources are the same distance from this point, P.

$$p_1 = A\cos\omega_1 t$$

$$p_2 = A\cos\omega_2 t$$

$$p = p_1 + p_2 = A(\cos\omega_1 t + \cos\omega_2 t)$$

$$p = 2A\cos\left(\frac{\omega_1 - \omega_2}{2}t\right)\cos\left(\frac{\omega_1 + \omega_2}{2}t\right)$$

We also note the average value of omega as $\omega_{\text{avg}} = (\omega_1 + \omega_2)/2$ and the variation from this value as $\Delta\omega = (\omega_1 - \omega_2)/2$

$$p = 2A\cos(\Delta\omega t)\cos(\omega_{\text{avg}} t)$$

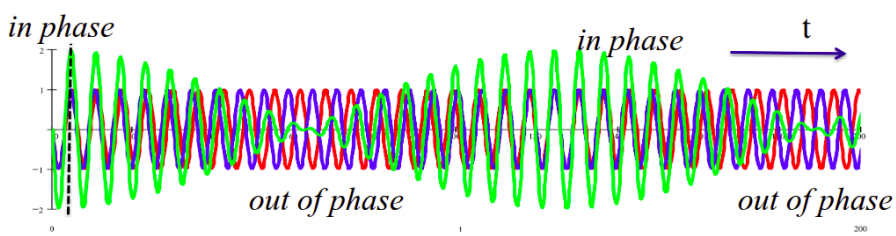


Figure 7.2: The resultant wave p in green, with p_1 and p_2 in red and blue.

Note that (as the difference in frequency is small) the two waves start mostly in phase, leading to a high amplitude resultant wave. As they drift out of phase, the resultant amplitude drops (becoming zero when in antiphase). Phase difference is periodic, so they return in phase and this repeats, creating a cosine wave with oscillating amplitude.

We can consider the 'envelope' that contains the resultant wave (given in purple):

The amplitude oscillates with angular frequency:

$$\omega = \left(\frac{\omega_1 - \omega_2}{2}\right) = 2\pi f \implies f = \left(\frac{\omega_1 - \omega_2}{4\pi}\right)$$

The human ear cannot really tell the difference between the green signal and the purple envelope signal, and we have a perceived beat frequency of:

$$f_{\text{beat}} = 2\left(\frac{\omega_1 - \omega_2}{4\pi}\right) = \frac{\omega_1}{2\pi} - \frac{\omega_2}{2\pi} = |f_1 - f_2|$$

The 2 arises as our ears don't care about the difference between a positive and a negative amplitude. The cosine wave goes positive, zero, negative, zero (one oscillation) while the human ear perceives loud, quiet, loud, quiet (two oscillations).

Note: This holds up when the amplitude of the two waves is the same. If the amplitudes are different, we would have to re-derive to take this into account.

Thu 23 Oct 2025 13:00

Lecture 8 - EM Standing Waves and Lasers

1 Lasers

A traditional laser has two mirrors on either side of a cavity. One is (ideally) 100% reflecting, while another is almost perfectly reflecting (say 99% reflective), but allows some transmission. This causes EM waves to reflect back and forth (with some transmitted to actually cause the visible laser).

These reflecting waves cause a standing wave inside the laser cavity. This standing wave oscillates with the equation given in Lec 07.

2 A Quick Audio Interlude

Listening to various sounds from a frequency generator, we notice two things:

- The human ear is very sensitive to changes in frequency, even in the single Hz range.
- A square wave has a higher percieved pitch than a sine wave of the same frequency.
  - This is because a sqaure wave can be decomposed into the sum of many sine waves, and some of these sine wave components have a higher frequency compared to the square wave itself, which out ears can detect.

3 And Back to Superposition

Given two travelling waves:

$$y_1 = A\cos(k_1 x - \omega_1 x)$$
$$y_2 = A\cos(k_2 x - \omega_2 x)$$
$$y = y_1 + y_2$$

Hence:

$$y = A\cos(k_1 x - \omega_1 x) + A\cos(k_2 x - \omega_2 t)$$

And using a double angle formula:

$$y = 2A\cos\left(\frac{k_1 - k_2}{2}x - \frac{\omega_1 - \omega_2}{2}t\right)\cos\left(\frac{k_1 + k_2}{2}x - \frac{\omega_1 + \omega_2}{2}t\right)$$

Or briefly:

$$y = 2A\underbrace{\cos(\Delta k x - \Delta\omega t)}_{\text{group}}\underbrace{\cos(k_{avg}x - \omega_{avg}t)}_{\text{carrier}}$$

We define the phase velocity as:

$$v_{\text{phase}} = \frac{\omega_{avg}}{k_{avg}}$$

And the group velocity as:

$$v_{\text{group}} = \frac{\Delta\omega}{\Delta k}$$

4 Carriers and Groups

What actually is a carrier and a group? Lets take radio as an example. Radio waves travel for potentially hundreds of kilometers, and it's not possible to send sound waves remotely close to that distance because of high attenuation.

The idea therefore arose of using a radio frequency wave to carry sound. Lets consider an example EM RF wave at $500kHz$. We then approximate our sound wave as a sine wave of pressure variations, say at $1000Hz$.

The first thing we can do is to modulate the amplitude of the RF wave by the audio wave. The RF wave keeps the same frequency, but changes amplitude depending on the amplitude of the audio wave.

Upon receipt of the wave, we filter out the high frequency oscillations of the radio wave itself, but we extract a lower frequency wave defined by a wave that passes through the maximum points of oscillation. This effectively restores the sound wave.

This is known as AM radio. The carrier is the high frequency wave, and the group is the lower frequency constructed wave.

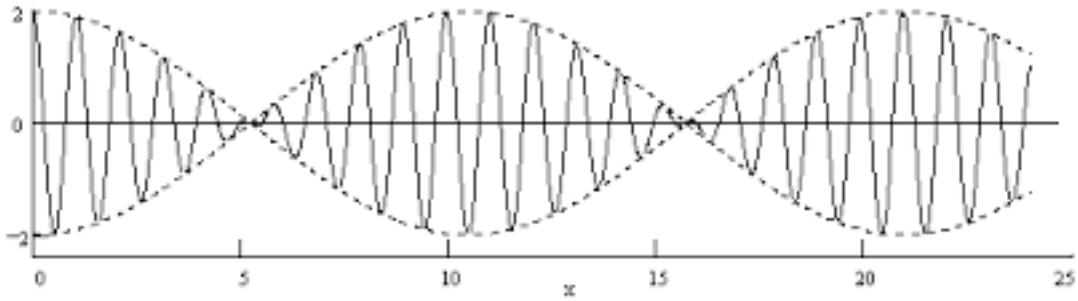


Figure 8.1: Carrier and Group waves. The lower freq. dotted line is the extracted wave (group), the higher frequency wave is the carrier.

5 Velocities

Which of the two waves have a higher speed?

Assuming wave speed and frequency are independent, we have:

$$\boxed{V = \omega_1/k_1 \qquad V = \omega_2/k_2}$$

$$v_{\text{phase}} = \frac{\omega_{\text{avg}}}{k_{\text{avg}}} = \frac{(\omega_1 + \omega_2)/2}{(k_1 + k_2)/2} = V$$

And for group:

$$v_g = \frac{(\omega_1 - \omega_2)/2}{(k_1 - k_2)/2} = V$$

Hence they move at the same speed. This is only true on the assumption that wave speed does not depend on frequency, which is not generally true. Depending on the circumstances, phase velocity can be greater or smaller than group velocity.

In a medium where the wave speed depends on frequency, we call this medium dispersive.

Air is non-dispersive (i.e. sound), and in a vacuum the same applies (i.e. speed of light).

What about the speed of light in glass? Does this depend on frequency? We'll investigate later in the optics section.

6 A Few More Definitions

Coherence: Two (or more) sources that are in phase, or have a constant phase difference (Δ) are said to be coherent.

6.1 Adding Coherent Waves

Constant phase difference means we can nicely add amplitudes, so total intensity:

$$I = |A|^2 = (A_1 + A_2 + A_2, \dots, A_n)^2$$

6.2 Adding Incoherent Waves

We directly add intensities instead:

$$I = I_1 + I_2 + \dots = A_1^2 + A_2^2 + \dots$$

6.3 The Decibel Scale

$$\beta = 10 \log\left(\frac{I}{I_0}\right)$$

Where I is the intensity of the source, and I_0 is a reference level approximately at threshold of hearing equal to $10^{-12} W/m^2$

At the threshold of hearing:

$$\beta = 10 \log\left(\frac{10^{-12}}{10^{-12}}\right) = 10 \log(1) = 0dB$$

Crucially, 0dB is not zero sound intensity.

At the upper pain threshold:

$$\beta = 10 \log\left(\frac{1}{10^{-12}}\right) = 120dB$$

But, how do we calculate I here?

$$I = \frac{\text{Power}}{\text{Area}}$$

Power is related to energy, so:

$$I = \frac{\text{energy}}{\text{time} \times \text{area}}$$

And to get intensity for an actual volume:

$$I = \frac{energy \times length}{time \times volume}$$

$$I = \frac{energy}{volume} \times wavespeed$$

What displacement on the eardrum do we actually have? For 120dB, 10^{-5}m, or 10^{-11} for 0dB. The latter measurement is smaller than an atom. The ear is a very sensitive instrument.

7 Human Senses

Interestingly, we see, hear and smell all on a log scale. Physical feeling is not very scientific and can't easily be quantified. Emotional feeling is also not quantifiable...

Tue 29 Oct 2025 12:00

Lecture 9 - Doppler Shift, Shockwaves and Optics I

Doppler Shift: If a wave emitter and observer are moving relative to each other, the observed frequency will be different to the emitted frequency. As wave emitting objects moves towards an observer, the perceived frequency of the wave increases. As the object moves away, the frequency decreases.

We consider two scenarios: (1) a moving source, and (2) a moving receiver, where the other is stationary. We will consider this for **sound waves**.

1 Case 1: Moving Source, Stationary Receiver

The source has frequency f_s and period $T_s = 1/f_s$. It has velocity u_s, and the velocity of waves in the medium is v.

At $t = 0$: The wave source creates a sound wave pulse, which begins to propagate out in all directions.

At $t = T_s$: The initial sound wave has moved by $d = st$, so distance travelled by the wave is vT_s distance units. The source has moved to the right by u_sT_s distance units.

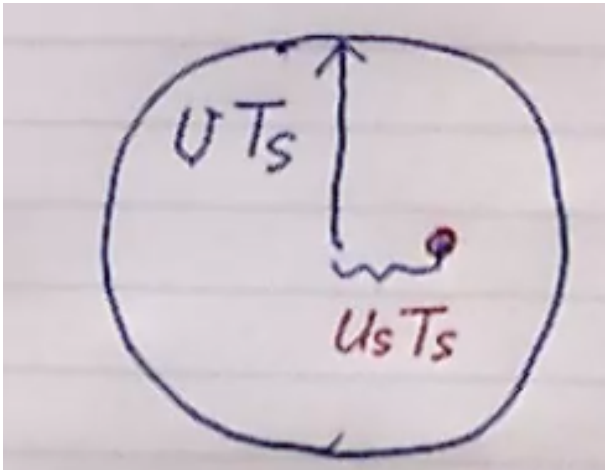


Figure 9.1

As this continues, we get something which looks like this:

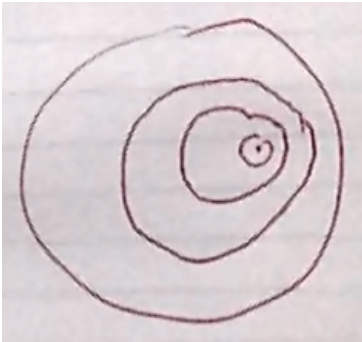


Figure 9.2

This effectively gives us a wave which has two different frequencies and two different wavelengths, one in the forward direction and one each in the backwards.

We denote the distance between the source's new location and the wavelet in the forward direction as λ_f, and in the reverse direction as λ_b.

$$\lambda_f = vT_s - u_sT_s$$

And the new frequency in the forward direction:

$$f_f = \frac{v}{\lambda_f}$$

$$f_f = \frac{v}{(v-u_s)T_s}$$

$$f_f = \frac{v}{(v-u_s)(1/f_s)}$$

$$f_f = f_s\frac{v}{v-u_s}$$

We can do the same analysis considering the wavelength in the backwards direction:

$$f_b = f_s\frac{v}{v+u_s}$$

2 Case 2: Stationary Source, Moving Receiver

Consider a periodic wave with wavelength λ and speed v. Lets look at the case where the receiver is moving towards the source at a speed u_r.

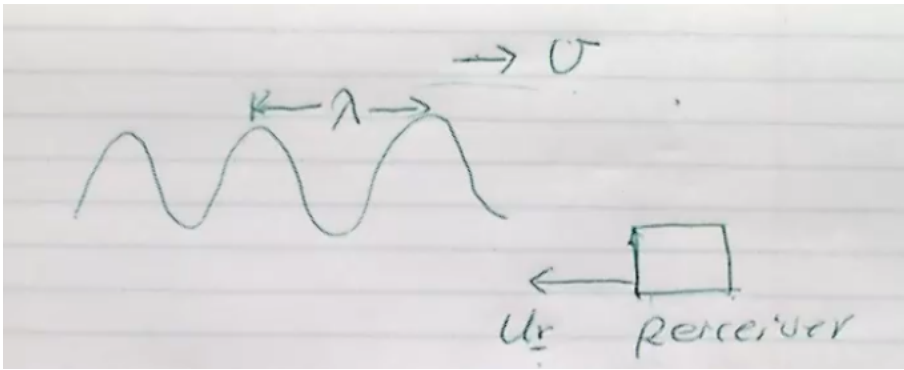


Figure 9.3

The receiver perceives the wave as moving at speed $v + u_r$ [1]. It therefore observes the frequency:

$$f_f = \frac{v+u_r}{\lambda}$$

[1] This does not take special relativity into account, so no ultrafast receivers here, and since it's a sound wave, $v \ll c$.

And if the receiver is moving in the opposite direction:

$$f_b = \frac{v - u_r}{\lambda}$$

Note that as the source is not moving, the wavelength remains constant.

3 Case 3: Both Source & Receiver Moving

In summary, for a moving source:

$$f_r = \frac{v}{v \pm u_s} f_s \quad (- \text{ towards, } + \text{ away.})$$

And for a moving receiver:

$$f_s = \frac{v \pm u_r}{v} f_r \quad (+ \text{ towards, } - \text{ away.})$$

We can combine these two equations to get the general form:

$$f_r = \frac{v \pm u_r}{v \pm u_s} f_s$$

Note that the two ±s are independent and the numerator/denominator values for the sign should be chosen according to the sign conventions for the numerator/denominator stated in the two individual equations. If moving towards, frequency increases. If moving away, frequency decreases.

3.1 What about light waves?

The doppler shift in frequency depends on the source/receiver speed relative to the medium. For light, we do not consider a traditional medium. We therefore cannot talk about absolute motion, we can only consider relative motion between the source and the receiver.

We have:

$$f_r = f_s \sqrt{\frac{c \pm u}{c \mp u}}$$

If we expand this and ignore terms in u^2/c^2 (as they'll be tiny). We get:

$$\frac{\Delta f}{f_s} = \pm \frac{u}{c} \qquad \text{where: } \Delta f = f_r - f_s$$

This only holds for $u \ll c$, as if this isn't true the u^2/c^2 terms will be non-negligible. The square root appears due to time dilation.

- **Blueshift (+):** Source approaching (frequency increases).
- **Redshift (−):** Source receding (frequency decreases).

4 Shock Waves

Shock waves arise when the velocity of the source is faster than the velocity of waves in the medium. We have previously implicitly assumed $u < v$, so this did not happen. We end up with a large number of wavefronts building up in front of the object, forming a cone of high amplitude:

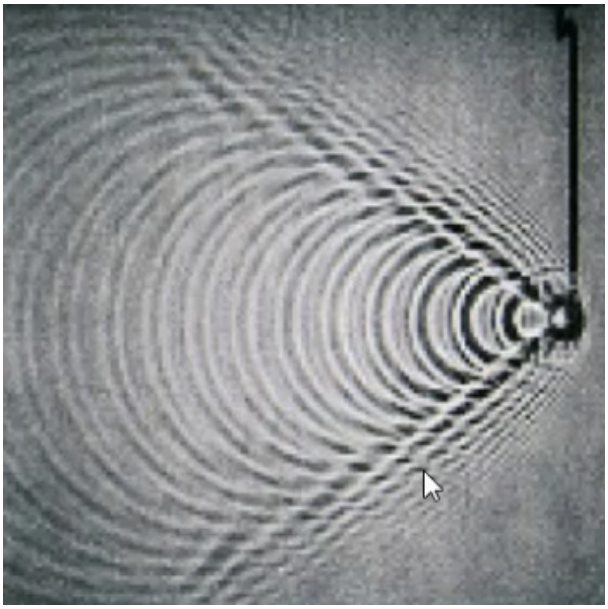


Figure 9.4: A shock wave cone.

If $u = v$, this creates a shock wave which is a straight line tangent to the motion.

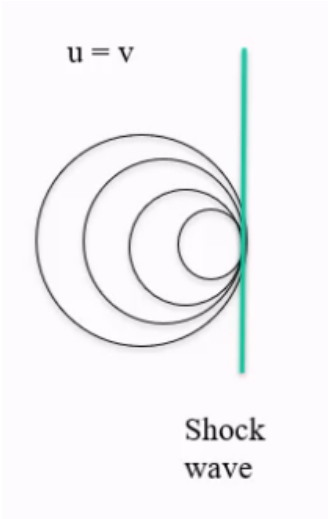


Figure 9.5

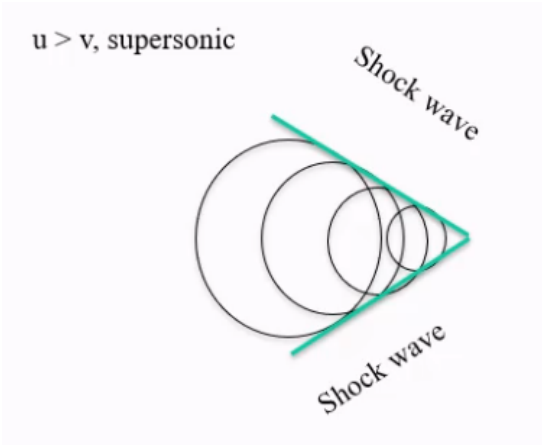


Figure 9.6

The angle created from the horizontal to either the top or bottom of this cone is known as the Mach Cone Angle θ, and is given by the Mach Number u/v (where v is velocity of the sound, u is the velocity of the source):

$$\sin\theta = \frac{v}{u}$$

When travelling along the path of the moving source, the sound waves produced later are heard first. This is the reverse of what we'd expect for a slower source. This is because, as the source moves faster than sound, the source gets closer to the receiver faster than the old sound waves can travel. the sound waves emitted at (for example) $t = 10s$ are therefore emitted much closer to the receiver and arrive before the sound emitted at (for example) $t = 5s$.

At the back of the supersonic source, the frequency is significantly shifted:

$$f_b = \frac{v}{v+u} f_s < \frac{1}{2} f_s$$

This equation suggests that in front of the source, the frequency becomes negative. This is because this analysis does not apply in front of the source, as waves do not pass the cone. The exception to this is a sonic boom, which is the extremely high amplitude of the cone itself being heard, as passes a receiver.

5 Optics Part One

Now we've studied waves, we can formally apply this to light. Electromagnetic waves are oscillations of the electric (E) and magnetic (B) fields.

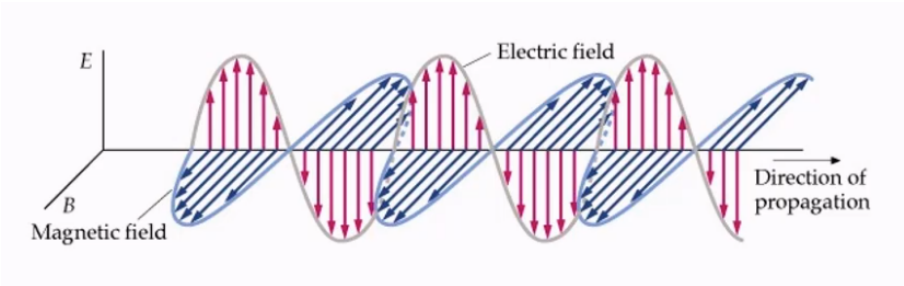


Figure 9.7

With the following properties:

- Light needs no medium, it can travel in a vacuum.
- Light does not involve the oscillation of particles (ignoring weird QM stuff).
- Light is a transverse wave (with f, ω, k, standing waves, doppler shift etc).

For electromagnetic waves, we have the wave equation:

$$\frac{\partial^2 E}{\partial x^2} = \epsilon_0 \mu_0 \frac{\partial^2 E}{\partial t^2}$$

Where ϵ and μ are two constants which determine how well waves propagate through electric and magnetic fields respectively. The zero subscript indicates that we are talking about free space (i.e. in a vacuum).

We have the following for the speed of light:

$$c = \sqrt{\frac{1}{\epsilon_0 \mu_0}}$$

Hence:

$$c^2 = \frac{1}{\epsilon_0 \mu_0} \implies 1/c^2 = \epsilon_0 \mu_0$$

Therefore the wave equation derived previously (with $1/v^2$ in place of $\epsilon_0 \mu_0$) is still satisfied.

For materials with *relative permittivity* ϵ_r and *relative permeability* μ_r (permittivity, ϵ and permeability μ for a specific material relative to the free-space values), the speed of light is:

$$v = \frac{c}{\sqrt{\epsilon_r \mu_r}} = \frac{c}{n}$$

Where $n = \sqrt{\epsilon_r \mu_r} > 1$. Light therefore travels slower in matter.

Thu 30 Oct 2025 13:00

Lecture 10 - Optics II

1 Key Principles

There are two main principles that we'll use:

1. **Huygen's Principle**
   - Each point on a wavefront serves as a source of spherical secondary *wavelets* that advance with speed and frequency identical to the primary wave.
   - If we consider each point on a wave as a spherical wavelet source, the wavefront is given by a line tangent to all of these.
2. **Fermat's Principle**
   - The actual path taken by a beam of light is the one which takes the least time to traverse.
   - i.e. $dt/dl = 0$
   - Light could theoretically take an infinite number of possible routes between any two points, but in practice (in the same material) it will travel in a straight line as this is the fastest route.
   - When light is travelling through an interface (i.e. an air to glass boundary), the fastest point is no longer a straight line, as refraction appears.

2 Reflection

Lets consider some mirror as a reflecting surface, with light travelling from point A to point B. There are multiple different theoretical paths that the light could take, so Fermat's principle requires the shortest possible time taken.

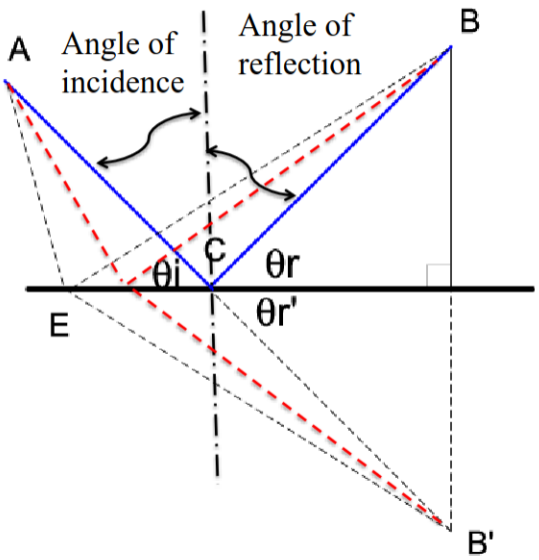


Figure 10.1

We require, by Fermat's Principle (as same material, so shortest time is simply shortest path):

$$AC + CB = \min$$

By considering congruent triangles (let O be the midpoint of B and B'), therefore:

$$AC + CB = AC + CB'$$

Provided ACB' is a straight line, $\theta_i = \theta_r (= \theta_{r'})$. Therefore the angles of incidence and reflection must be equal.

3 Refraction

When light is travelling through a material, the speed of light in each is different. Therefore, a straight line is no longer the fastest path.

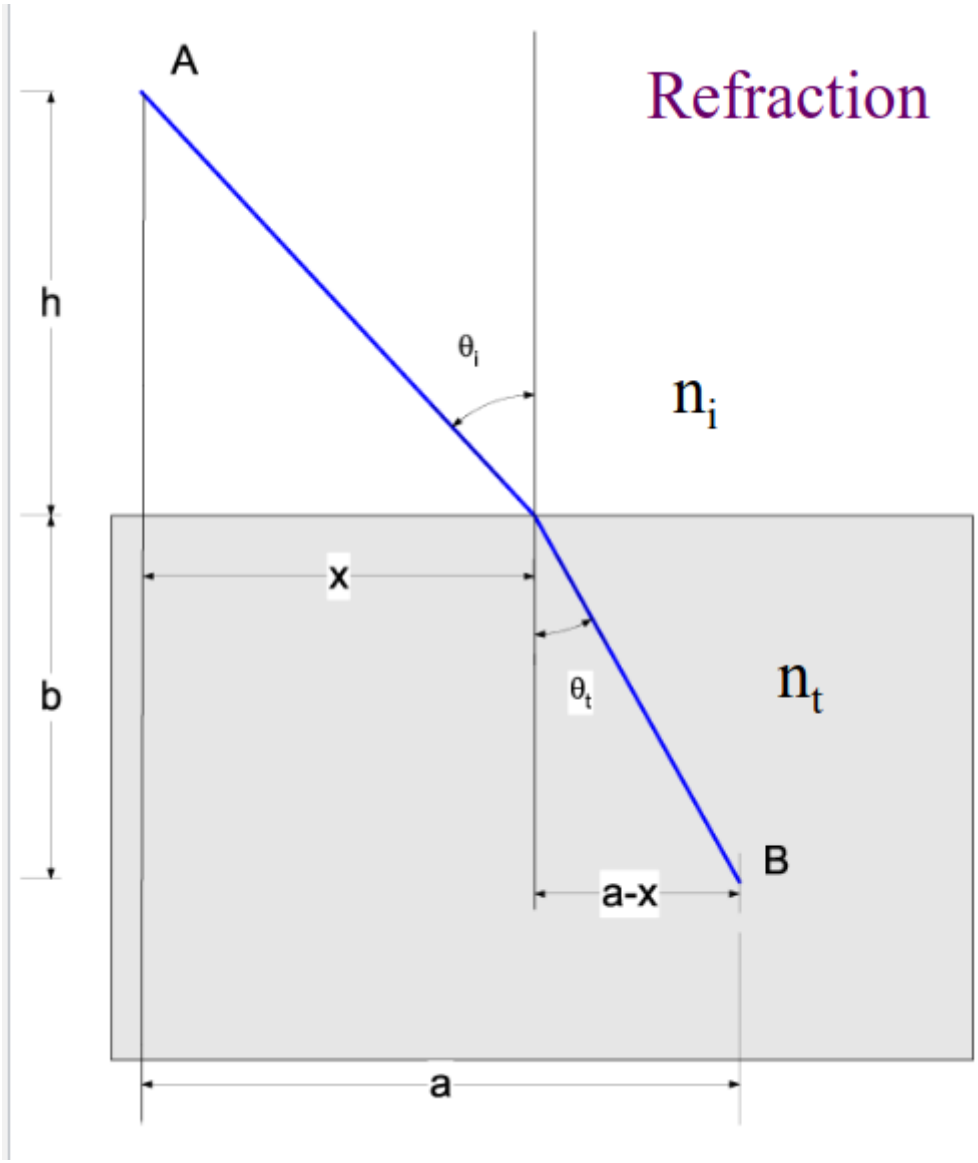


Figure 10.2

Where:

- n_i is the incident index.
- n_t is the post-refraction index.
- θ_i is the angle of incidence.

- θ_t is the angle of refraction.
- Light is travelling from point A to point B.
- Distances are given per diagram.

The aim is to be able to find the true trajectory (per Fermat) that takes the shortest amount of time. By s = d/t, for given distance values:

$$t = \frac{\sqrt{h^2 + x^2}}{v_i} + \frac{\sqrt{b^2 + (a - x)^2}}{v_t}$$

Where v_i, v_t are speeds of light in the respective media. t is a function of x, with constants, so to find a min value we differentiate wrt x and solve for minima:

$$\frac{dt}{dx} = \frac{x}{v_i\sqrt{h^2 + x^2}} + \frac{-(a - x)}{v_t\sqrt{b^2 + (a - x)^2}} = 0$$

Substituting values for distances, we get:

$$\frac{\sin\theta_i}{v_i} = \frac{\sin\theta_t}{v_t}$$

And since $n = c/v$, we get:

$$n_i \sin\theta_i = n_t \sin\theta_t \tag{10.1}$$

The shortest (wrt time) path must have angles which satisfies this. This is called Snell's Law.

4 Connecting Fermat and Huygen

Speed of light in a medium is less than in a vacuum, this is categorised by the index of refraction, n:

$$n = \frac{c}{v}$$

Where c is speed of light in a vacuum, and v is speed of light in the medium. i.e. for water, $n = 1.333$, for air, $n = 1.0003$.

Suppose we have a wavefront travelling as a series of wavelets. As soon as each wavelet hits the glass, they slow down. This leads to an uneven distribution of speeds across the wavelength, as some have slowed down and some have not. Therefore, the wave bends around and we get a new wavefront at a different angle to the previous wavefront.

Note the condition where the light travels perpendicular to the interface, i.e. $\theta_i = \theta_t = 0$. In this case, all wavelets hit the medium and change speed at exactly the same time, therefore there is no difference in velocity across the wavefront, so no direction change.

5 But how?

How does light actually "choose" which path to take? Apparently it just does and we have to wait to find out...

6 Combining Reflection and Refraction

When light strikes a boundary surface, there is two components - both reflected and transmitted (just like on a string). To determine this, we have to use the wave equation:

$$E = E_0 \cos(kx - \omega t)$$

Noting that instead of traditional mechanical amplitude we use the amplitude of the electromagnetic field at this point, E_0. For now, we ignore polarisation as a potential scenario.

7 More Index

We note that since $v = f\lambda$, the index of refraction will change depending on the frequency of light:

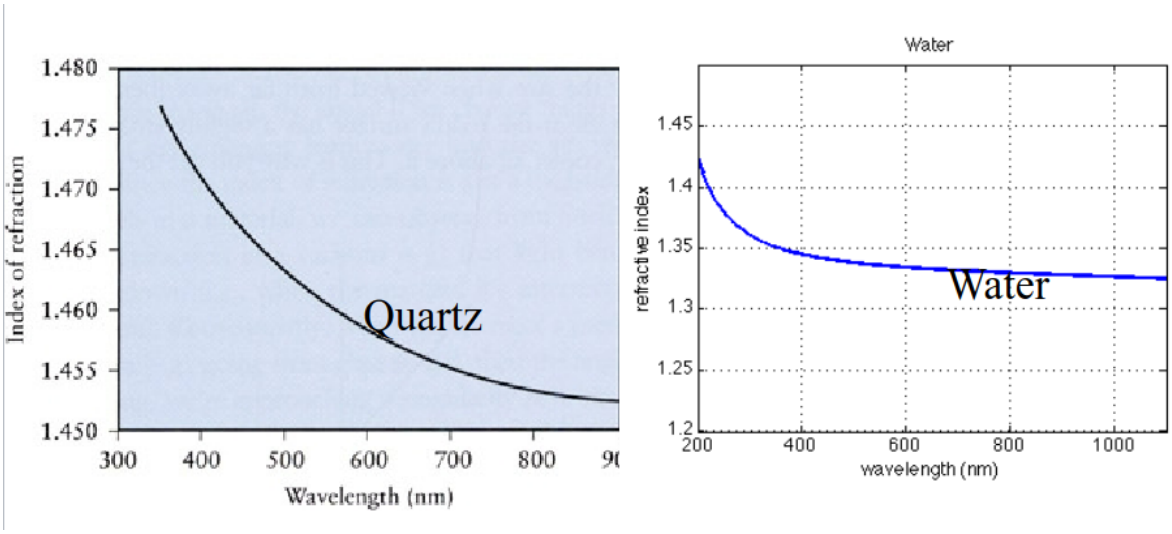


Figure 10.3

When we shine light of multiple frequencies (i.e. white light), the higher frequency light bends further than the lower frequency light. This causes dispersion of light, i.e. the formation of a rainbow out of a prism. This arises because each frequency has a different speed and therefore a different change in θ.

If we shine pulses of light of multiple frequencies down a fibre optic line, the higher frequencies will have a higher n, hence a lower speed and will arrive later.

This allows us to explain the formation of sunsets and rainbows etc. In the case of a sunset, when we see the sun just above the horizon, the sun has actually set just below the horizon. We cannot therefore see the sun via direct line of sight, and yet we can still see it as if we could?

This is because the light from the sun is refracted and bends towards us. The atmosphere is higher density at the bottom, and lower density as altitude increases. The amount of refraction depends on density, hence changing the index of refraction. We can say that n is a function of y, where y is height.

We imagine it as being actually present precisely where we see it, because our brain does not account for this refraction and extrapolates the light as a straight line. We therefore see the sun has just about to set, when in reality the sun physically has set below the horizon.

Wed 05 Nov 2025 11:00

Lecture 11 - Reflection and Refraction Examples I

1 Simple Mirror

A man is standing in front of a simple flat mirror. He is height hm. What is the minimum height of the mirror if he wishes to see the full image of himself?

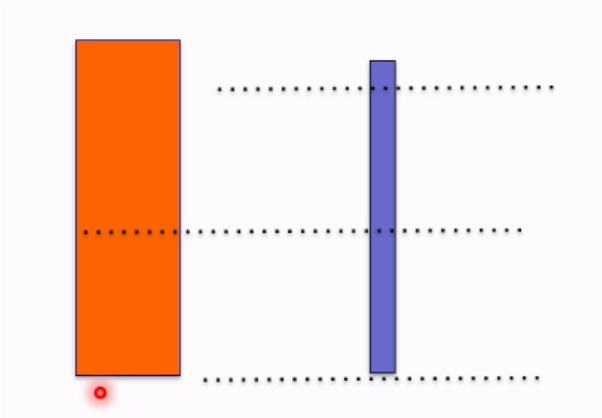


Figure 11.1: The man, modelled as a box (orange) and the mirror of indeterminate hight (blue)

Consider a ray from the top of the man's head to his eye level:

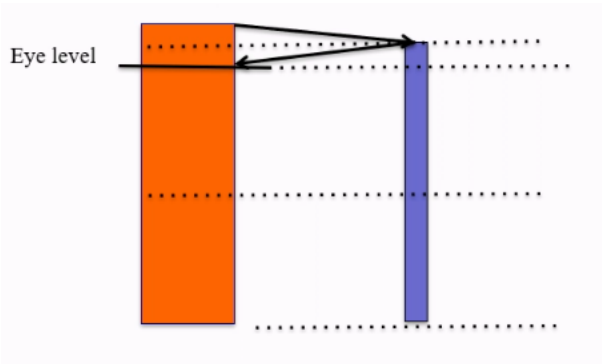


Figure 11.2

And a ray from feet level again to the man's eye (we also label some dimensions and add two extra red rays):

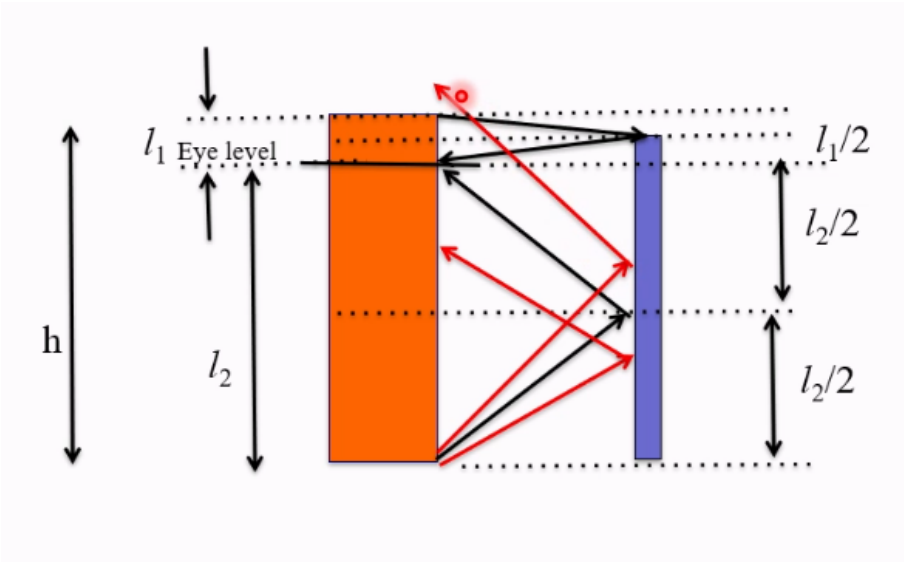


Figure 11.3

Note that the red rays are not observed by the man. The lower red ray strikes the man at too low an angle so is below eye level, while the opposite is true for the upper red ray. Neither are therefore seen.

This means that the lower half of the mirror is redundant, as any rays which strike the lower section will be reflected at too low an angle to meet the man's eyes. The upper portion of the mirror is therefore the only required portion, giving a final height of:

$$l_1/2 + l_2/2 = (l_1 + l_2)/2 = h/2$$

2 Corner Cube Reflector

Note: Pg 1088 Y&F A corner cube reflector is three mutually orthogonal reflecting surfaces:

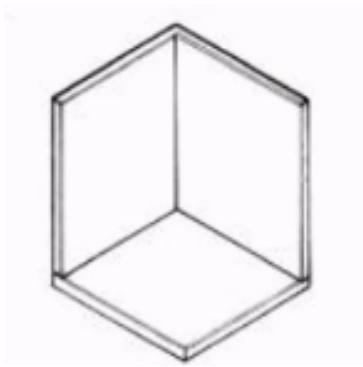


Figure 11.4: Corner Cube Reflector

We'll consider a simplified setup, with two orthogonal mirrors in a 2D scenario:

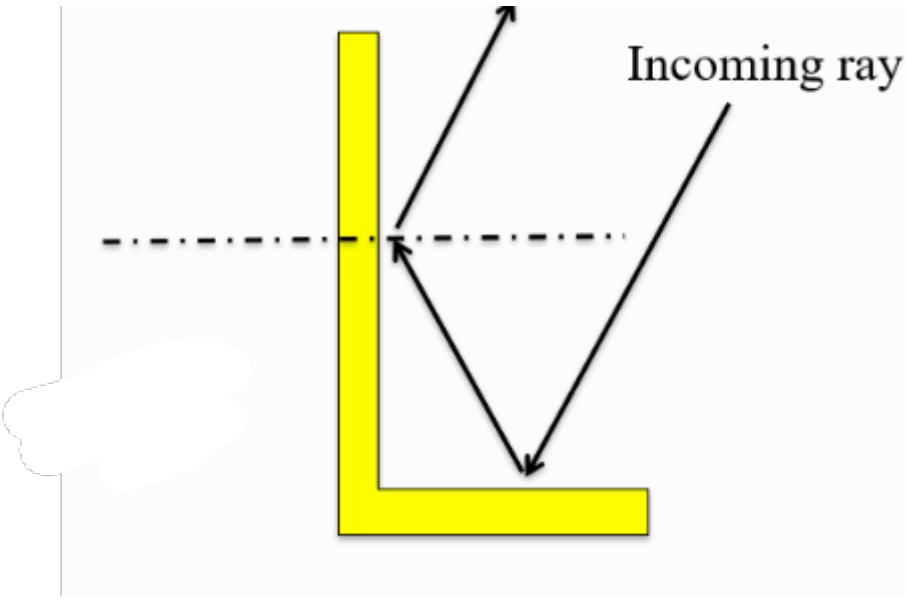


Figure 11.5

Here, the incoming ray is reflected twice (assuming it does not arrive parallel to a mirror), once on each mirror. We can easily analyse the problem as we know that the angle of incidence and reflection for the initial incoming ray are equal, denoted θ_1.

The angle of incidence and reflection for the second reflection are also equal, and are given by $90 - \theta_1$. Note that this is equal to the angle made by the incoming wave with the horizontal.

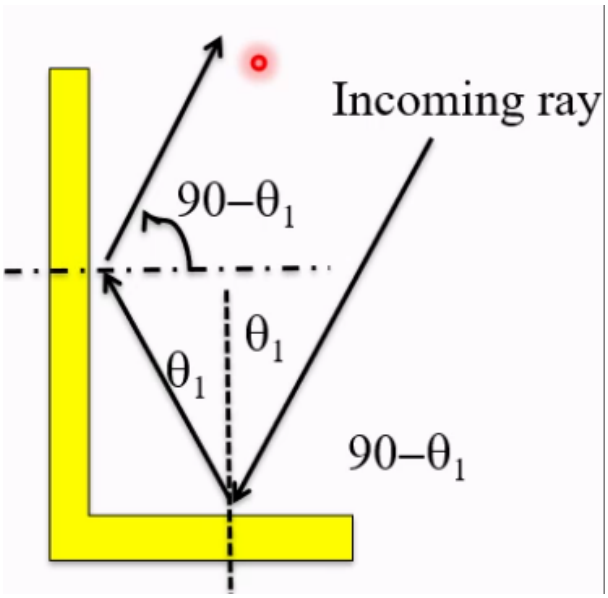


Figure 11.6

This means that the reflected ray travels back towards the emitter, at the same angle as it was initially.

3 Light Travelling Through a Piece of Glass

Consider a 1cm thick slab of glass, where a light ray is fired into it at angle of incidence θ_1.

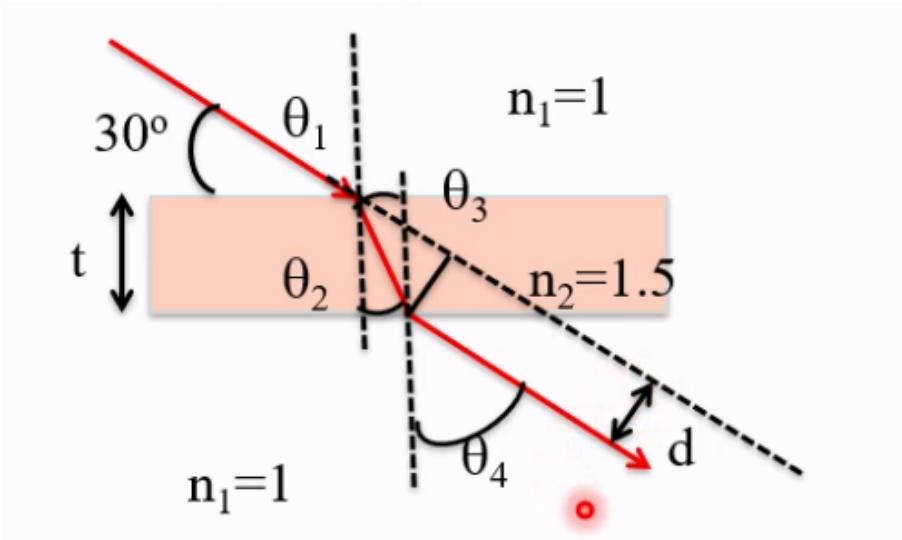


Figure 11.7

We use Snell's Law to find theta two:

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$
$$\theta_2 = \arcsin\left(\frac{n_1 \sin\theta_1}{n_2}\right) = 35.3°$$

And again:

$$n_2 \sin\theta_3 = n_1 \sin\theta_4$$
$$\theta_2 = \theta_3$$
$$\implies \theta_4 = \arcsin\frac{n_2 \sin\theta_3}{n_1} = 60°$$

Therefore as $\theta_1 = \theta_4$ the light ray exits the material a the same angle it entered, just with some displacement. We denote this displacement d. We can find it again using some trig:

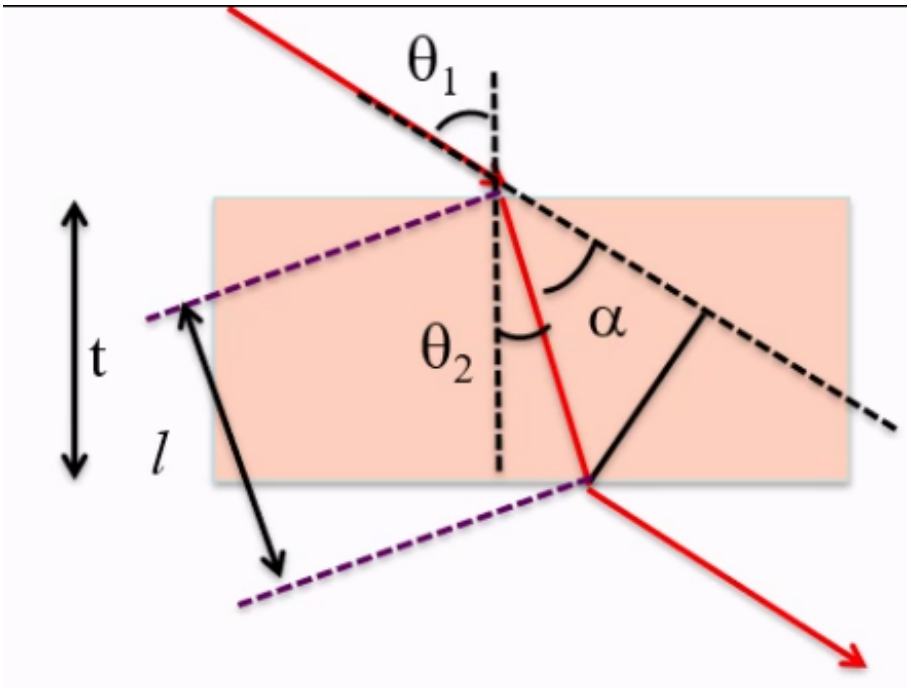


Figure 11.8

The distance is given by:

$$d = l \sin\alpha = l \sin(\theta_1 - \theta_2)$$
$$= \frac{t}{\cos\theta_2} \sin(\theta_1 - \theta_2) = 0.51\text{cm}$$

4 Apparent Depth of an Object

Consider an example with a fish in the pond. We want to estimate how

Thu 06 Nov 2025 13:00

Lecture 12 - Reflection and Refraction Examples II

Wed 12 Nov 2025 11:00

Lecture 13 - Image Formation

1 Sign Rules

We have the mirror equation:

$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f}$$

The numerical values of s and s' give us the distance, but the position of the formed image is determined by their signs.

2 Mirrors

The image formed by a mirror is:

- *Virtual*, as we effectively see it appears as being behind the mirror, where there is no real light.
- *Erect*, as there is no inversion (if you stand in front of a mirror, you see your body in the same top-to-bottom orientation as real life).
- *Reversed*, as if you look in a mirror your body faces you, and you see your face, not your back.

2.1 Angled Mirrors

If we have a pair of hinged mirrors, changing the angle between them changes the way that the image is produced. Crucially, it changes how many images of the object we see:

Figure 13.1